

Timothy Mitchell

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EDUCATION

University of North Carolina at Charlotte, Charlotte, NC

Bachelor's of Science in Mechanical Engineering,

Expected Graduation December 2027

Durham Tech Community College, Durham, NC

Associates in Engineering,

May 2025

RELATED COURSEWORK:

Statics | Solids | Mechatronics 1 | Manufacturing Systems | Machine Shop | Dynamics | Thermodynamics |
Mechatronics 2 | Linear Algebra

SKILLS

Solid Works, Onshape, Python, Java, React, C, C++, Visual Studio Code, Excel, Microsoft Tools, Computer Hardware, 3D modeling, 3D design, Manufacturing, 3D printing, Electronics, KiCAD

PROJECTS

Technical Lead, Rival Robotics Competition Team, Raleigh, NC <https://ncrivals.com/> August 2024 – Present

- Designing a 3D model using Onshape CAD of a competition robot based on criteria and material constraints
- Implementing Manufacturability and machinability into our robot design
- Developed a Website using React to display the team's accomplishments and information
- Wrote Blog posts for the team describing our robot and its technical information
- Competed in California for competitions against teams all over the US

Team Lead, Spark Combat Robotics, Durham, NC

August 2023 – Present

- Developing 2 Combat Robots using CAD, going through several iterations and versions
- Selected optimized materials and construction techniques, improving robot durability and performance in competition.

Python Platforming Game, Durham, NC

January 2024 – Present

- Programmed from Scratch using Python, a 2D platformer game using advanced classes and libraries
- Implemented a custom recursive generation algorithm to develop terrain and different levels automatically
- Uses object-oriented programming to make game expansion easier

ENGINEERING EXPERIENCES

Formula Student Vehicle Dynamics Design Team Member, 49ers Racing, Charlotte, NC January 2026 – Present

- Working with a team of 40+ students to develop a formula-style race car from scratch every 18 months
- Developed Vehicle dynamic simulations to verify suspension kinematics
- Used SolidWorks to CAD parts and used Ansys to verify strength and optimize topology

Manufacturing Engineering Intern, Northeast Tool and Manufacturing, Mathews NC

August 2025 – Present

- Developing and CADing fixtures for CNC Lathe and Mill parts
- Worked on several continuous improvement projects, resulting in successful implementation on time

Lead Mentor, Codewiz FTC Team, Durham, NC

August 2023 – August 2025

- Lead a team of 10 students to build a robot in 4 months, competing in competitions
- Taught advanced CAD principles and Java Programming to students
- Organised grants and received over \$2000 worth of funding for 2 years of team expenses

Co-Team Captain, FRC Team 587 the Hedgehogs, Hillsborough, NC

August 2019 – June 2022

- Managed a team of over 30 students to back-to-back state championship wins
- Learned CAD skills to be able to design and fabricate a 120lb robot
- Lead the Electrical team to develop and integrate a well-organized and robust system of electronics

OTHER EXPERIENCE

Center Director / Manager

August 2023 – August 2025

Code Wiz Durham, Durham, North Carolina

- Led and trained a team of 10 coaches, managing scheduling and performance for team success.
- Organized and Ran school partnerships teaching robotics in classroom settings of 15+ kids